Answer ____________________ cm [2]

21 **(a)** Simplify $\left(\dfrac{4a^6}{b^4}\right)^{-\frac{1}{2}}$, giving your answer in positive index.

Answer ______________________________ [2]

(b) $\dfrac{2^k}{\sqrt[4]{8}} = 4^{2k}$.

Use the laws of indices to find the value of k.
Show your working.

Answer $k =$ ___________________________ [3]

22 The first three terms in a sequence of numbers are given below.

$$T_1 = 1 \times 2 + 10 = 12$$
$$T_2 = 2 \times 3 + 6 = 12$$
$$T_3 = 3 \times 4 + 2 = 14$$
$$T_4 = 4 \times 5 - 2 = 18$$

(a) Show that the n^{th} term of the sequence, T_n, is given by $T_n = n^2 - 3n + 14$.

Answer

[2]

(b) Evaluate T_{50}.

Answer ____________________ [1]

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(c) Explain why every term in the sequence is even.

Answer ____

____

____

____ [2]